

Lab Session 5 — Specification

Introduction to Applied Regression Analysis

ICPSR Summer Program 2026

This lab uses data from Koch, Michael T., and Sarah A. Fulton. “In the Defense of Women: Gender, Office Holding, and National Security Policy in Established Democracies.” *The Journal of Politics* 73, no. 1 (2011): 1–16. It will focus on different model specifications. Each row of the dataset is a country-year. The column `ccode` gives its Correlates of War country code. This is the original data from the article, so there are a lot of variables to contend with. For now, focus on `mil_exp_percdp2` (military spending per capita), `m_femlead` (Female Leader), and `m_woman2` (Percentage of women in parliament). Examine these data closely using the techniques we reviewed.

```
## You will need the haven library to import Stata data files
library(haven)
women12 <- read_dta("women12.dta")
```

1. Explore the dataset
2. Fit a basic linear regression model

```
model1 <- lm(mil_exp_percdp2 ~ m_woman2 + m_femlead + cap
             + mad_gdppc + postcold2, data = women12)
```

3. Fit a model with an interaction term between `m_femlead` (Woman Leader) and `m_woman2` (Percentage of Women in Parliament).
4. Interpret the coefficients of the interaction model
5. Plot the interaction effect of `m_woman2` and `m_femlead`.
6. Assess the assumptions of the interaction model.
7. Fit a log-linear model
8. Interpret the coefficients of the log-linear model
9. Fit a polynomial model (quadratic)
10. Plot the polynomial model